

Fundamentals of Electrical and Electronics Engineering (FEEE)

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Lecture-18

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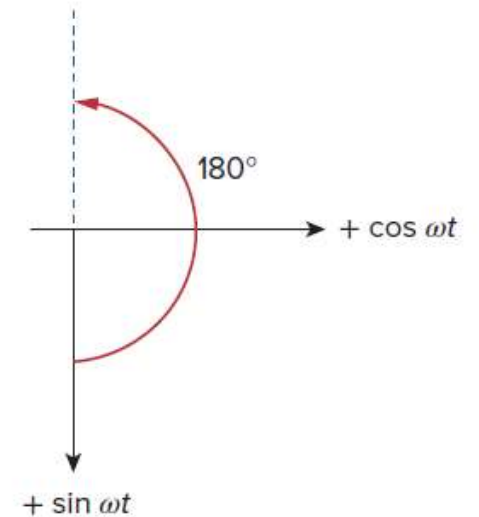
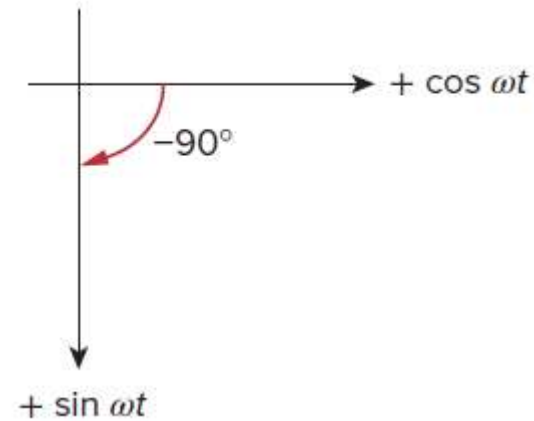
IV. (a) Transform phasors to sinusoids (b) Transform sinusoids to phasors

$$v(t) = V_m \cos(\omega t + \phi) \quad \Leftrightarrow \quad \mathbf{V} = V_m \angle \phi$$

(Time-domain representation) (Phasor-domain representation)

Sinusoid-phasor transformation.

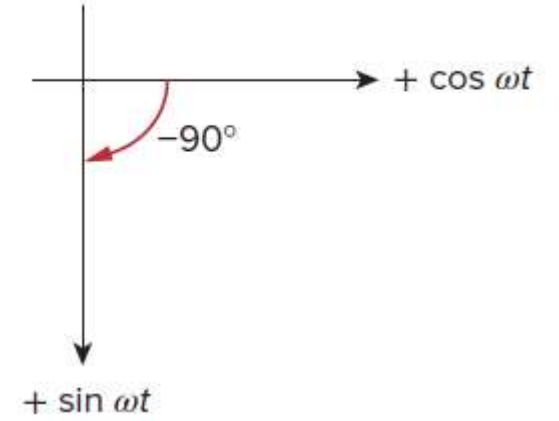
Time domain representation	Phasor domain representation
$V_m \cos(\omega t + \phi)$	$V_m \angle \phi$
$V_m \sin(\omega t + \phi)$	$V_m \angle \phi - 90^\circ$
$I_m \cos(\omega t + \theta)$	$I_m \angle \theta$
$I_m \sin(\omega t + \theta)$	$I_m \angle \theta - 90^\circ$



1.Transform sinusoids to phasors

(a) $i = 6 \cos(50t - 40^\circ) \text{ A}$

(b) $v = -4 \sin(30t + 50^\circ) \text{ V}$



2. Express these sinusoids as phasors:

(a) $v = -14 \sin(5t - 22^\circ) \text{ V}$

(b) $i = -8 \cos(16t + 15^\circ) \text{ A}$

3. Find the sinusoids represented by these phasors:

(a) $\mathbf{I} = -3 + j4$ A

(b) $\mathbf{V} = j8e^{-j20^\circ}$ V

4. Find the sinusoids corresponding to these phasors:

(b) $\mathbf{I} = j(12 - j5) \text{ A}$

5. Given $i_1(t) = 4 \cos(\omega t + 30^\circ)$ A and $i_2(t) = 5 \sin(\omega t - 20^\circ)$ A, find their sum.

V. Phasor Relationships for Circuit Elements

1. Voltage-current relations for a Resistor

- If the current through a resistor R is $i = I_m \cos(\omega t + \phi)$, the voltage across it is given by Ohm's law as

$$v = iR = RI_m \cos(\omega t + \phi)$$

- The phasor form of this voltage is

$$\mathbf{V} = RI_m \angle \phi$$

But the phasor representation of the current is $\mathbf{I} = I_m \angle \phi$. Hence,

$$\mathbf{V} = R\mathbf{I}$$

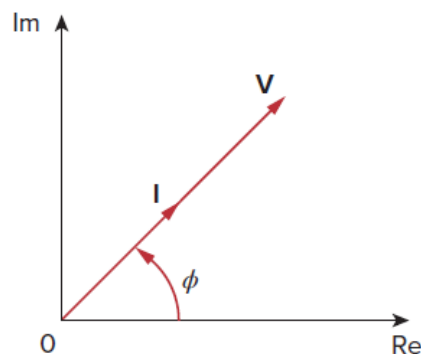
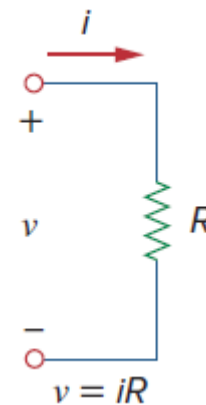
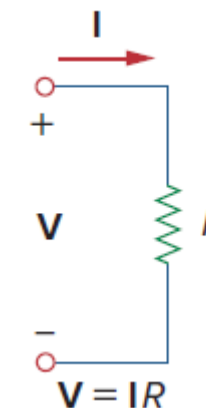


Fig: Phasor diagram for the resistor- voltage and current are in phase



Voltage-current relations for a resistor in the: (a) time domain,



(b) frequency domain.

2. Voltage-Current relations for an inductor

- For the inductor L , assume the current through it is $i = I_m \cos(\omega t + \phi)$. The voltage across the inductor is

$$v = L \frac{di}{dt} = -\omega L I_m \sin(\omega t + \phi)$$

$$v = \omega L I_m \cos(\omega t + \phi + 90^\circ)$$

- which transforms to the phasor

$$\mathbf{V} = \omega L I_m e^{j(\phi + 90^\circ)} = \omega L I_m e^{j\phi} e^{j90^\circ} = \omega L I_m \angle \phi + 90^\circ$$

But $I_m \angle \phi = \mathbf{I}$,

$$e^{j90^\circ} = j.$$

$$\mathbf{V} = j\omega L \mathbf{I}$$

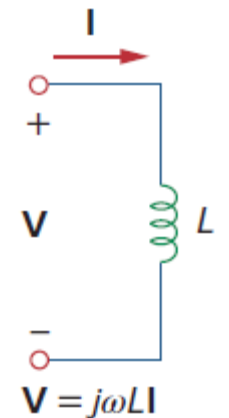
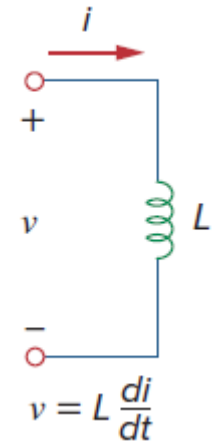
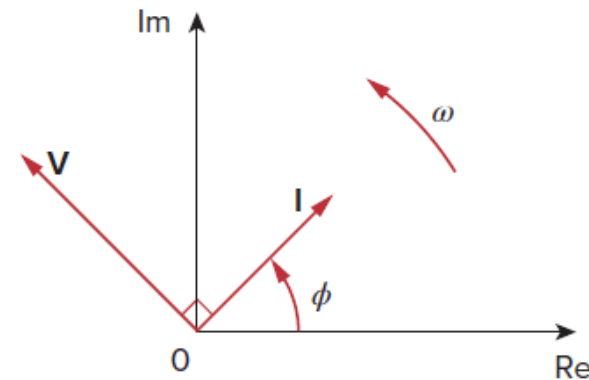


Fig: Phasor diagram for the inductor; $\mathbf{I}$ lags $\mathbf{V}$.

3. Voltage-Current relations for a Capacitor

- For the capacitor C , assume the voltage across it is $V = v_m \cos(\omega t + \phi)$. The current through the capacitor is

$$i = C \frac{dv}{dt}$$

$$I = j\omega CV \Rightarrow V = \frac{I}{j\omega C}$$

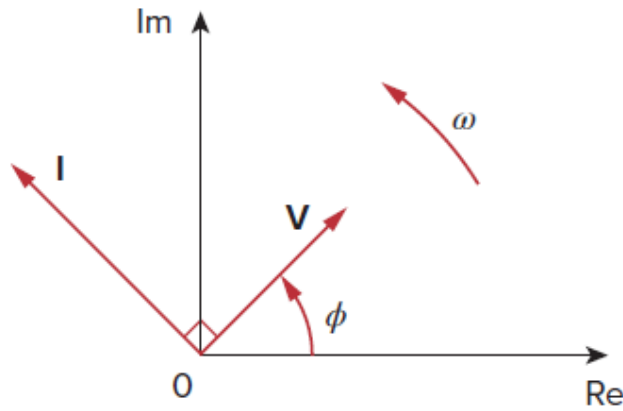


Fig: Phasor diagram for the capacitor; I leads V .

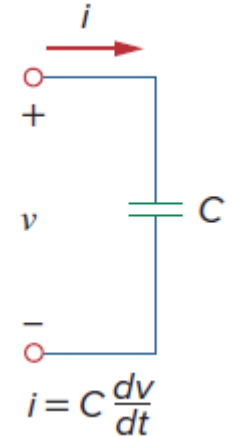
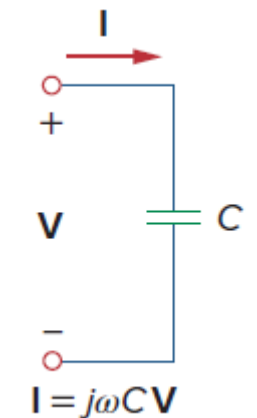


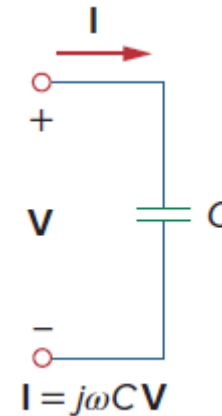
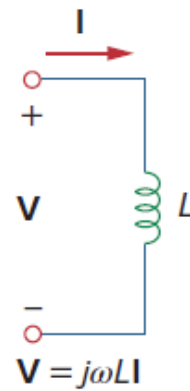
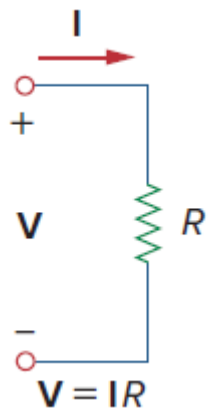
Fig: Voltage-current relations for a capacitor in the: (a) time domain



(b) Frequency domain.

Summary of voltage-current relationships.

Element	Time domain	Frequency domain
R	$v = Ri$	$V = RI$
L	$v = L \frac{di}{dt}$	$V = j\omega LI$
C	$i = C \frac{dv}{dt}$	$V = \frac{I}{j\omega C}$



VI. (a) Impedance

- In the preceding section, we obtained the voltage-current relations for the three passive elements as

$$V = RI, \quad V = j\omega LI, \quad V = \frac{I}{j\omega C}$$

$$\frac{V}{I} = R, \quad \frac{V}{I} = j\omega L, \quad \frac{V}{I} = \frac{1}{j\omega C}$$

$$\boxed{Z = \frac{V}{I} \quad \text{or} \quad V = ZI}$$

Impedances and admittances of passive elements.

Element	Impedance	Admittance
R	$Z = R$	$Y = \frac{1}{R}$
L	$Z = j\omega L$	$Y = \frac{1}{j\omega L}$
C	$Z = \frac{1}{j\omega C}$	$Y = j\omega C$

- where Z is a frequency-dependent quantity known as *impedance*, measured in ohms.

- As a complex quantity, the impedance may be expressed in rectangular form as

$$Z = R \pm jX$$

$$Z = |Z| \angle \theta$$

$$Z = R \pm jX = |Z| \angle \theta$$

$$|Z| = \sqrt{R^2 + X^2}, \quad \theta = \tan^{-1} \frac{\pm X}{R}$$

$$R = |Z| \cos \theta, \quad X = |Z| \sin \theta$$

Impedances and admittances of passive elements.

Element	Impedance	Admittance
R	$Z = R$	$Y = \frac{1}{R}$
L	$Z = j\omega L$	$Y = \frac{1}{j\omega L}$
C	$Z = \frac{1}{j\omega C}$	$Y = j\omega C$

(b) Admittance

- The **admittance** Y is the reciprocal of impedance, measured in siemens (S).

$$Y = \frac{1}{Z} = \frac{I}{V}$$

$$Y = G + jB$$

P-1: Find the input impedance of the circuit in Fig. below. Assume that given circuit operates at $\omega = 50$ rad/s.

Z_1 = Impedance of the 2-mF capacitor

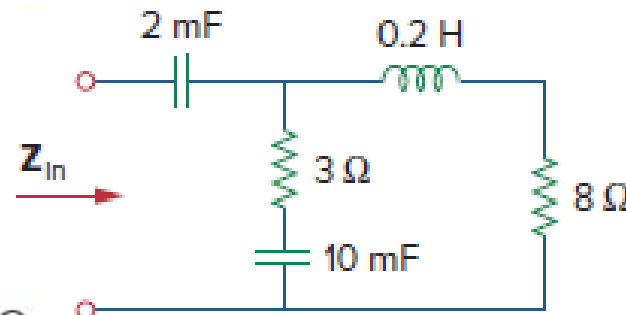
Z_2 = Impedance of the 3- Ω resistor in series with the 10-mF capacitor

Z_3 = Impedance of the 0.2-H inductor in series with the 8- Ω resistor

$$Z_1 = \frac{1}{j\omega C} = \frac{1}{j50 \times 2 \times 10^{-3}} = -j10 \Omega$$

$$Z_2 = 3 + \frac{1}{j\omega C} = 3 + \frac{1}{j50 \times 10 \times 10^{-3}} = (3 - j2) \Omega$$

$$Z_3 = 8 + j\omega L = 8 + j50 \times 0.2 = (8 + j10) \Omega$$

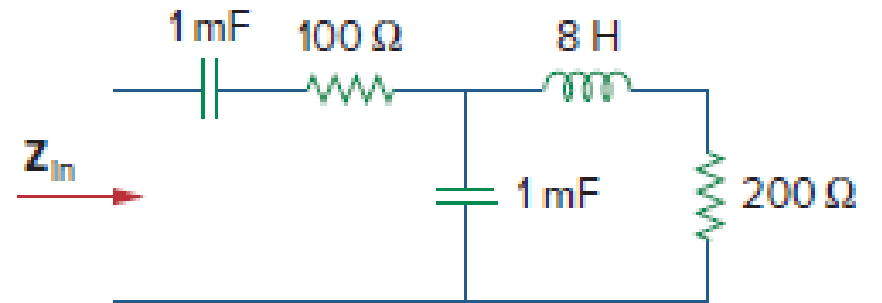


$$\begin{aligned} Z_{in} &= Z_1 + Z_2 \parallel Z_3 \\ &= -j10 + \frac{(3 - j2)(8 + j10)}{11 + j8} \\ &= -j10 + \frac{(44 + j14)(11 - j8)}{11^2 + 8^2} \end{aligned}$$

$$= -j10 + 3.22 - j1.07 \Omega$$

$$Z_{in} = 3.22 - j11.07 \Omega$$

2. Determine the input impedance of the circuit in Fig. at $\omega = 20$ rad/s.



$$Z_{in} = (109.53 - j105.29)\ \Omega.$$

3. Find Z_{in} ?

$$10 \text{ mF} \Rightarrow \frac{1}{j\omega C} = \frac{1}{j4 \times 10 \times 10^{-3}} = -j25 \Omega$$

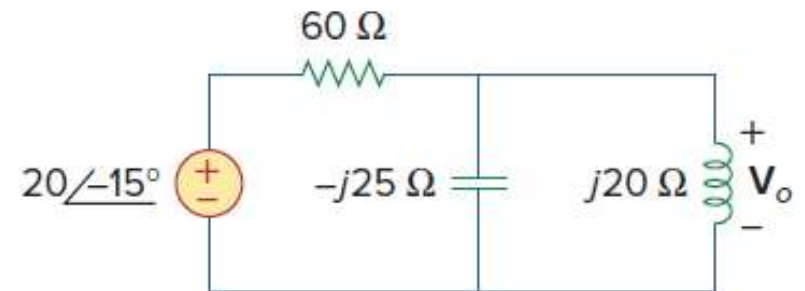
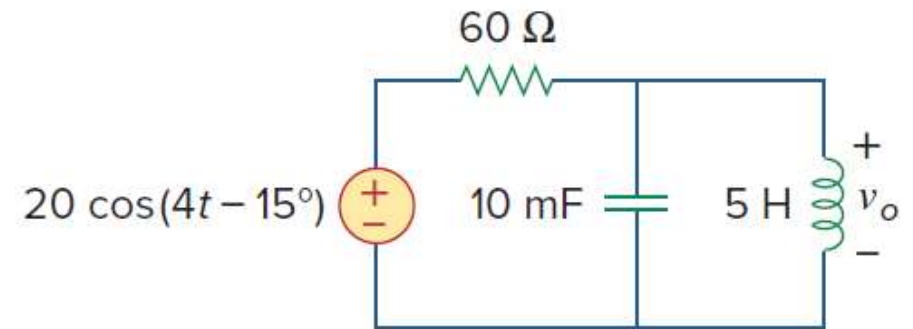
$$5 \text{ H} \Rightarrow j\omega L = j4 \times 5 = j20 \Omega$$

Z_1 = Impedance of the 60- Ω resistor

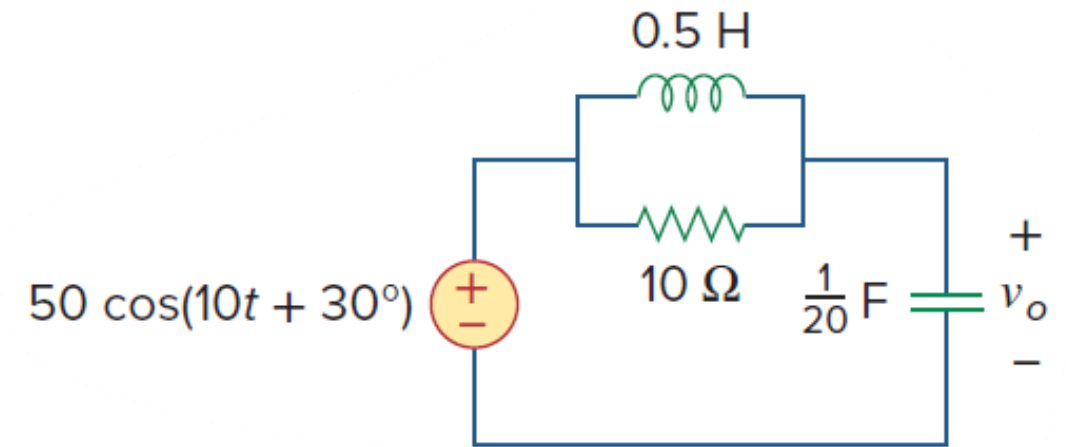
Z_2 = Impedance of the parallel combination of the 10-mF capacitor and the 5-H inductor

Then $Z_1 = 60 \Omega$ and

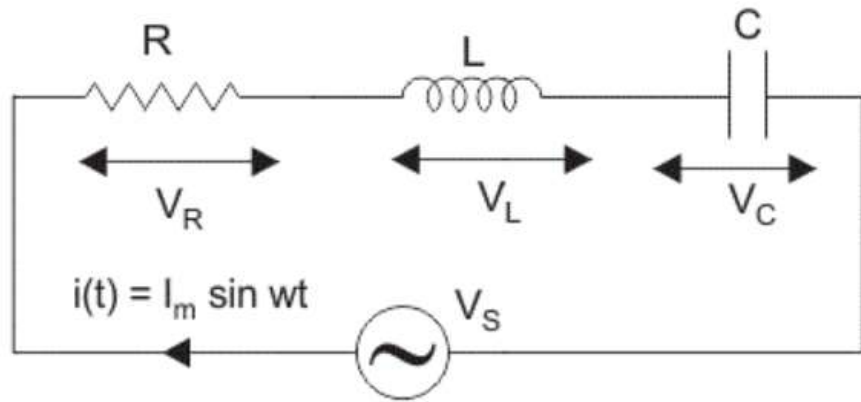
$$Z_2 = -j25 \parallel j20 = \frac{-j25 \times j20}{-j25 + j20} = j100 \Omega$$



4. Find Z_{in} ?



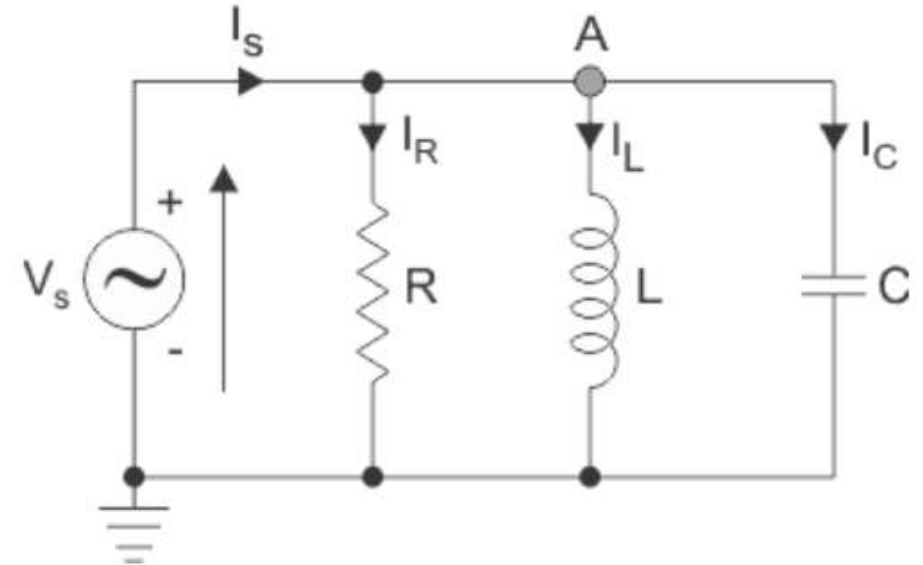
5 (a) Series RLC



Where $V_R = IR$, $V_L = IX_L$, $V_C = IX_C$

$$\text{impedance } Z = \sqrt{R^2 + (X_L - X_C)^2}$$

(b) Parallel RLC



$$I_R = \frac{V}{R}, \quad I_C = \frac{V}{X_C}, \quad \text{and} \quad I_L = \frac{V}{X_L}$$

$$\text{Admittance } \frac{1}{Z} = Y = \sqrt{\left(\frac{1}{R}\right)^2 + \left(\frac{1}{X_L} - \frac{1}{X_C}\right)^2}$$